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13.29) Perturbation field: $\hat{H}' = 2\hat{H}'(\vec{r}) \cos \omega t$

Let it act for a long time, and let $\omega \approx \omega_{kl}$.

Since we are dealing with absorption,

$$P_{lk} = \frac{4|\hat{H}'_{kl}|^2}{\hbar^2(\omega_{kl} - \omega)^2} \sin^2\left[\frac{1}{2}(\omega_{kl} - \omega)t\right]$$

Since $\hat{H}'$ acts for a long time, taking the limit $t \rightarrow \infty$ we have

$$P_{lk} \rightarrow \frac{2\pi t}{\hbar^2} |\hat{H}'_{kl}|^2 \delta(\omega_{kl} - \omega)$$

Thus the δ -function indicates that the only state to be excited is the kl state.

The Fourier decomposition of $\hat{H}'$ in time:

$$\begin{aligned} \int_{-\infty}^{\infty} \hat{H}' e^{i\omega t} dt &= 2\hat{H}'(\vec{r}) \int \cos \omega_0 t e^{i\omega t} dt \\ &= \hat{H}'(\vec{r}) \int (e^{i\omega_0 t} + e^{-i\omega_0 t}) e^{i\omega t} dt \\ &= \hat{H}'(\vec{r}) \int (e^{i(\omega_0 + \omega)t} + e^{i(\omega - \omega_0)t}) dt \end{aligned}$$

Recall $\int_{-\infty}^{\infty} e^{i(a+b)t} dt = 2\pi \delta(a+b)$

$$\Rightarrow = 2\pi \hat{H}'(\vec{r}) (\delta(\omega + \omega_0) + \delta(\omega - \omega_0)),$$

Picking out $\omega_0 \rightarrow -\omega$ or $\omega_0 \rightarrow \omega$.

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13.33 1-D Harmonic Oscillator. (charge to mass q/m , spring constant K in its ground state.

We have $E(t) = \mathcal{E} \cos \omega_0 t$. Thus $V(t) = -e\mathcal{E}\hat{x} \cos \omega_0 t$.

We write $\hat{H}' = -e\mathcal{E}\hat{x} \cos(\omega_0 t)$
 $= \hat{H}'(x) \mathcal{L} \cos(\omega_0 t) \Rightarrow \hat{H}'(x) = -e\mathcal{E}\hat{x}$

Since $(\omega_{n0} - \omega_0)t \ll 1$ use

$$P_{0n} = \frac{t^2 |\hat{H}'_{n0}|^2}{\hbar^2}. \text{ Now, } H_{n0} = -e\mathcal{E} \langle n | x | 0 \rangle$$

$$= -e\mathcal{E} \sqrt{\frac{\hbar}{2m\omega_0}} \langle n | \hat{a} + \hat{a}^\dagger | 0 \rangle$$

$$= -e\mathcal{E} \sqrt{\frac{\hbar}{2m\omega_0}} \langle n | 1 \rangle \quad \uparrow \delta_{n,1}$$

Thus

$$P_{0n} = \delta_{n,1} \left| \frac{e\mathcal{E}}{\sqrt{2} \hbar \beta} \right|^2 t^2 \quad \text{with } \beta^2 = \frac{m\omega_0}{\hbar}$$

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Shankar 18.2.1

$$C_0(t) = \frac{-i}{\hbar} \int_{-\infty}^{\infty} dt e^{i\omega_0 t} (-e\mathcal{E}) \langle 1 | \hat{x} | 0 \rangle \frac{1}{1 + (\frac{t}{\tau})^2}$$

$= \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger)$

Where $\omega_0 = \frac{E_1^{(0)} - E_0^{(0)}}{\hbar} = \hbar\omega \frac{\frac{3}{2} - \frac{1}{2}}{\hbar} = \omega$

$$C_0(t) = \frac{-i}{\hbar} \sqrt{\frac{\hbar}{2m\omega}} (-e\mathcal{E}) \int_{-\infty}^{\infty} dt e^{i\omega t} \frac{1}{1 + (\frac{t}{\tau})^2}$$

$$= \frac{+ie\mathcal{E}\tau^2}{\sqrt{2m\omega\hbar}} \int_{-\infty}^{\infty} dt \frac{e^{i\omega t}}{t^2 + \tau^2}$$

Using integral table: $= \frac{ie\mathcal{E}\tau^2}{\sqrt{2m\omega\hbar}} \frac{\pi}{\tau} e^{-\omega\tau}$

$$P = |C|^2 = \frac{e^2 \mathcal{E}^2 \tau^2}{2m\omega\hbar} e^{-2\omega\tau} \quad \text{QED}$$